

This is a **hedging engine**, not a “try to guess the market” system. I’ll walk it back exactly with your numbers, but in my own structure, and I’ll keep repeating:

⚠ **This is NOT financial advice.**

Everything below is **educational**, just a mathematical way to understand how a prop-account + personal hedge-account structure could be engineered.

Real trading is risky. You can lose money on both sides.

1 Core idea – percentage loss vs dollar gain

1.1 Two accounts, opposite roles

You are running **two accounts in opposition**:

1. Prop account (evaluation → funded)

- Large *virtual* balance: e.g. **\$200,000**
- Rules: **max daily drawdown, max overall drawdown, profit targets**
- Measured/restricted in **percent (%)** of that big 200k.

2. Personal hedge account

- Your own **real money**
- Takes the **opposite side** of the prop trades
- Measured in **dollars (\$)** and actual P/L.

Every time the prop account **loses**, the hedge should **win**.

Every time the prop account **wins big**, the hedge should **lose in a controlled way**.

1.2 The key asymmetry

- A **small % move on a big 200k account** can be turned into a **meaningful \$ gain** on your much smaller personal account **if** the hedge size is chosen correctly.
- You are **deliberately shaping**:
 - The **“failure” branch**:
 - Evaluation fail → hedge profits enough to **refund the fee + extra**
 - Funded fail → hedge profits enough to **recover all costs + extra**
 - The **“success” branch**:
 - Evaluation pass → you accept a **known dollar cost** to obtain a funded account
 - Funded payout → prop payout is **much larger** than hedge loss

The whole system is about **engineering both branches** (win and lose on prop) to be **positive or controlled** over many cycles.

Again: this is a **mathematical model**, not a guarantee or a recommendation.

2 Evaluation logic with the 200k account

We use your **standard example** for everything:

2.1 Prop evaluation parameters

- **Account size:** \$200,000
- **Phase 1 target:** 10% = **\$20,000**
- **Phase 2 target:** 5% = **\$10,000**
- **Max daily drawdown:** 5% = **\$10,000**
- **Max overall drawdown:** 10% = **\$20,000**
- **Evaluation fee (challenge cost):** **\$1,000**

2.2 Your evaluation risk profile

You chose:

- **Risk per day (eval):** 2% of prop balance
- **Trades per day (eval):** 2
- **So risk per trade (eval):** 1% of prop balance

On a 200k eval:

- **1% per trade** = $0.01 \times 200,000 = \$2,000$ per trade
- **2 trades/day** → total risk per day = $2 \times 2,000 = \$4,000/\text{day}$

This sits safely under the **5% daily DD**, which is:

- **5% of 200k** = $0.05 \times 200,000 = \$10,000$

So your structure:

- **Max daily allowed loss:** \$10,000
 - **Planned daily risk:** \$4,000
 - You're not maxing out the rule; you're staying within it.
-

3 Hedge sizing for evaluation fail

Goal for a **full eval fail** (using the full 10% DD):

If I blow the evaluation (hit -10%),
I want the hedge to:

- Recover the **\$1,000 evaluation fee**, and
- Give me an additional **\$1,000 net profit**.

So:

- **Target net on evaluation fail (personal hedge):** +\$1,000
- **Plus eval fee:** +\$1,000
- **Total gross needed from hedge on a full 10% drawdown:**
= 1,000 (desired profit) + 1,000 (fee)
= **\$2,000 gross** from hedge

We spread that \$2,000 across the **10% max DD**.

3.1 Hedge yield per 1% prop loss (evaluation)

Formula:

$$\text{Hedge per 1\% prop loss (eval)} = \frac{\text{eval fee} + \text{desired profit on fail}}{\text{max DD in \%}}$$

Plug in your numbers:

- Eval fee = 1,000
- Desired profit on fail = 1,000
- Max DD = 10%

$$= \frac{1,000 + 1,000}{10} = \frac{2,000}{10} = \mathbf{\$200 \text{ per 1\% prop loss}}$$

Interpretation:

- Every **1% loss on the prop account** (which is **\$2,000 loss**) should correspond to **+\$200 profit** on the hedge.

Because:

- Risk per trade on prop = 1%, each **full SL** on prop is **-1% = -\$2,000**.

So you are designing:

Each full SL on prop $\approx$ +\$200 profit on the hedge.

4 Evaluation – maximum SLs and hedge result

4.1 Max number of SLs before evaluation breach

- **Max overall DD:** 10%
- **Risk per trade:** 1%

Maximum full-loss trades before breach:

$$\text{maxEvalSLTrades} = \left\lfloor \frac{10\%}{1\%} \right\rfloor = \left\lfloor 10 \right\rfloor = 10 \text{ trades}$$
$$\left\lceil \frac{10\%}{1\%} \right\rceil = \left\lceil 10 \right\rceil = 10 \text{ trades}$$

So:

- If you take **10 full SL trades** in a row on the prop:
 - Prop: $-10\% = -0.10 \times 200,000 = \textbf{-\$20,000}$ (evaluation failed)

4.2 Hedge profit per SL trade

We already fixed:

- **Hedge per 1% prop loss = \$200**

One SL = 1% loss on prop, so:

- **Per-SL profit on hedge = \$200**

Over **10 SL trades**:

$$\text{Gross hedge profit} = 10 \times 200 = \$2,000$$
$$\text{Gross hedge profit} = 10 \times 200 = \textbf{\$2,000}$$

Now subtract the eval fee:

- Gross hedge profit: **\$2,000**
- Eval fee paid: **\$1,000**
- Net result if you **blow the challenge**:

$2,000 - 1,000 = \$1,000$ net profit
 $2,000 - 1,000 = \mathbf{\$1,000}$ net profit

So if the evaluation **fails completely** (uses the full 10% DD):

- **Prop**: $-\$20,000$ (virtual) → challenge failed
- **Hedge**: $+\$2,000$
- **After fee**: $+\$1,000$ net in your pocket

! In this structure, the “**disaster**” case (**blowing the eval**) is *not* a disaster. It's **monetised**.

Again: this is theoretical and assumes perfect execution, fills, and mirrored hedging – real markets are messier.

5 One-trade pass for Phase 1 & Phase 2

Now you design the **success branch**: passing each phase with **one big trade**.

5.1 Phase 1 as a 10R target

- **Phase 1 target**: 10% of 200k = **\$20,000**
- **Risk per trade**: 1% = **\$2,000**

So the **R-multiple** of the Phase 1 win:

Phase 1 R-multiple = $\frac{10\%}{1\%} = 10R$
 $\mathbf{\text{Phase 1 R-multiple} = \frac{10\%}{1\%} = 10R}$

You also have the rule:

- Each **1R loss** on prop (1% loss) = **\$200 gain** on hedge (in the fail case).

- Symmetrically, a **1R gain** on prop (1% gain) implies a **\$200 loss** on hedge, if the hedge size is mirrored.

So if you hit the big trade and **Phase 1 passes in one shot**:

- Prop: **+10% = +\$20,000 → Phase 1 passed**
- Hedge: **$-10R \times \$200 = -10 \times 200 = -\$2,000$**

So:

Phase 1 personal cost $\approx$ \$2,000

(this is the hedge loss associated with that winning trade)

5.2 Phase 2 as a 5R target

Phase 2 parameters:

- **Target:** 5% of 200k = **\$10,000**
- **Risk per trade:** still 1% = **\$2,000**

So:

Phase 2 R-multiple = $\frac{5\%}{1\%} = 5R$

We keep the same mapping:

- **Per-1R move** on prop $\leftrightarrow$ **\$200 hedge P/L**

So a 5R winning trade on prop (to hit the 5% target):

- Prop: **+5% = +\$10,000 → Phase 2 passed**
- Hedge: **$-5R \times \$200 = -5 \times 200 = -\$1,000$**

So:

Phase 2 personal cost $\approx$ \$1,000 (hedge loss to pass).

6 Total cost to get funded

Now combine everything needed to go from **no account** to **funded**.

6.1 Hedge cost of passing both phases

- Phase 1 hedge cost: $\approx \$2,000$
- Phase 2 hedge cost: $\approx \$1,000$

Total hedge cost:

$$2,000 + 1,000 = 3,000 \quad 2,000 + 1,000 = \mathbf{3,000}$$

6.2 Add evaluation fee

Add the **challenge fee**:

- Eval fee: **\$1,000**

Total “**cost to get funded**” (your term):

$$\text{Total cost to get funded} = 3,000 + 1,000 = 4,000 \quad \text{Total cost to get funded} = 3,000 + 1,000 = \mathbf{4,000}$$

So if you **pass both phases cleanly with one trade per phase**:

- You have **paid ~ \$4,000 in total**:
 - \$1,000 eval fee
 - \$3,000 in hedge losses across the two winning trades

And in exchange:

- You now have a **\$200k funded account**, ready to run your **funded-stage hedging and payout engine**.
-

7 Funded hedge design & failure logic

Now you are **funded**:

- Funded account size: **\$200,000**
- Max overall DD still: **10% = \$20,000**

You set your funded trading risk profile to match:

- **Risk per day (funded): 2%**
- **Trades per day (funded): 2**
- So **risk per trade (funded): 1%** (same 1% structure)

7.1 What you want the hedge to do at funded fail

Your target for a **funded blow-up**:

If the funded account hits max DD (−10%),
the hedge should recover:

- the **full cost to get funded (4,000)**, and
- **\$1,000 extra profit** on that funded fail.

So:

- **costToFunded: \$4,000**
- **failFundedProfit target: \$1,000**
- **Total gross hedge gain desired on funded blow-up:**

$$4,000 + 1,000 = 5,000 \quad 4,000 + 1,000 = \mathbf{5,000}$$

Spread that across the same **10% max DD**.

7.2 Hedge per 1% loss on funded account

Formula:

$$\text{Hedge per 1\% prop loss (funded)} = \frac{\text{costToFunded} + \text{failFundedProfit}}{\text{max DD in \%}}$$

Plug in:

- $\text{costToFunded} = 4,000$
- $\text{failFundedProfit} = 1,000$
- $\text{max DD} = 10\%$

$$= \frac{4,000 + 1,000}{10} = \frac{5,000}{10} = \mathbf{\$500 \text{ per 1\% prop loss}}$$

Interpretation:

- For each **1% loss** on the funded prop account (which is $-\$2,000$ on prop),
- You want the hedge to **earn +\$500**.

Because:

- **Risk per trade (funded) = 1%**
- Each full SL = **-1% prop = $-\$2,000$**

So:

Per SL on funded account $\rightarrow$ hedge should profit +\$500.

7.3 Max funded SL trades

As before:

- **Max DD: 10%**
- **Risk per trade: 1%**

$$\text{maxFundedSLTrades} = \left\lfloor \frac{10\%}{1\%} \right\rfloor = 10 \text{ trades} \quad \text{maxFundedSLTrades} = \left\lfloor \frac{10\%}{1\%} \right\rfloor = 10 \text{ trades}$$

If the funded account takes **10 full SL trades** and hits the -10% limit:

- Prop: $-10\% = -\$20,000 \rightarrow$ funded account **failed**
- Hedge: **$10 \times \$500 = \$5,000$** gross profit

Now compare that to the sunk cost:

- $\text{costToFunded} = \$4,000$

Net after recovering the 4,000:

$$5,000 - 4,000 = 1,000 \quad 5,000 - 4,000 = \mathbf{1,000}$$

So in the **funded fail branch**:

- You **recover all prior costs** (eval fee + hedge to get funded)
- And you still walk away with **~\$1,000 net profit**

Again: in this model, **blowing the funded account** is not terminal — it is **monetised** via the hedge.

This is the same structural idea as the evaluation fail, but with **bigger dollar flows**: the hedge offers a “**parachute**” for the entire funded cycle.

8 Payout math and expected value per funded cycle

Now we look at the **success branch** of a funded cycle: **hitting a payout**.

8.1 Funded payout target

You pick a **payout target** of **5%** on the funded account.

On a 200k funded prop account:

- $5\% = 0.05 \times 200,000 = \$10,000$ gain in the prop balance.

Assume the prop firm gives a **90% profit split**:

- **Payout to you:**
 $= 10,000 \times 0.9 = \$9,000$

8.2 Risk multiple for the payout trade

Your funded risk structure:

- Risk per trade = **1%** of prop = **\$2,000**
- Payout target = **5%**

So the payout trade corresponds to:

$$\text{Payout R-multiple} = \frac{5\%}{1\%} = 5R$$
$$\text{Payout R-multiple} = \frac{1\%}{5\%} = 0.2R$$

8.3 Hedge loss when payout trade wins

We set:

- **Hedge per 1% prop loss on funded = \$500**

This maps similarly to R:

- **Per 1R move on prop** (1% move) $\leftrightarrow$ **\$500** of hedge P/L (profit when prop loses, loss when prop wins).

So if the **prop hits the 5% target** (a +5R win):

- Prop: +5% = **+\$10,000**
- Hedge: $-5R \times \$500 = -5 \times 500 = \textbf{-\$2,500}$

Now calculate your **net**:

- **Prop payout to you (90% split):** +\$9,000
- **Hedge loss on same move:** -\$2,500

$$\text{Net for payout event} = 9,000 - 2,500 = 6,500$$
$$\text{Net for payout event} = 9,000 - 2,500 = 6,500$$

So:

Each clean **payout hit** (from a flat start) is worth about **+\$6,500 net** (after paying for the hedge side of that trade).

8.4 Carry cost and multiple payouts

Conceptually:

- Initially, you had to **spend \$4,000** to get funded.
- But the **funded fail branch** is set up so that if you **eventually hit -10%**, your hedge recovers that \$4,000 plus \$1,000.

So over a **full long-term cycle (from scratch to eventual blow-up)**:

- Evaluation / recoup:
 - PASS path: pay ~4,000, but this is “covered” later by the funded fail hedge.
 - FAIL path: hedge gives +1,000 (eval refund + profit) and you can try again.
- Funded:
 - Every **payout hit** gives around **+\$6,500** net.
 - **Funded fail** later gives another **+1,000 net** after recouping the 4k.

So in this simplified model:

- **Payout branch** = big positive (~**\$6,500** per payout).
- **Fail branch** (eval or funded) = **0 + something positive**:
 - Eval fail = **+\$1,000**
 - Funded fail after recouping costs = **+\$1,000**

This is why you view it in **cycles** rather than individual trades.

9 Cycle overview – how the engine repeats

Think of it as two nested cycles:

1. **Evaluation cycle**
2. **Funded cycle**

9.1 Evaluation cycle

Start: You buy a **200k, 2-step evaluation**.

Two main paths within one eval attempt:

A. Evaluation fail (max DD hit)

- Prop:
 - Takes up to **10 SL trades** of -1% each $\rightarrow -10\% = -\$20,000$ virtual loss
- Hedge:
 - **Per 1% loss: +\$200**
 - Over 10% loss: $10 \times 200 = + \$2,000$

Net of eval fee:

- Hedge $+\$2,000 - \$1,000$ fee = **+\$1,000 net**

So a **failed evaluation** in this structure:

- **Refunds your fee + extra**, via the hedge.
- Leaves you **with more capital** to try again.

B. Evaluation pass with one trade per phase

- **Phase 1:**
 - Prop wins **$+10\% = +\$20,000$** (10R)
 - Hedge loses **$10 \times \$200 = -\$2,000$**
- **Phase 2:**
 - Prop wins **$+5\% = +\$10,000$** (5R)
 - Hedge loses **$5 \times \$200 = -\$1,000$**

Total hedge losses:

- $2,000 + 1,000 = \mathbf{\$3,000}$

Add eval fee:

- $3,000 + 1,000 = \mathbf{\$4,000}$

So:

You pay ~\$4,000 total to emerge with a **\$200k funded account**.

That \$4,000 is your “**costToFunded**”, which the funded hedge is explicitly designed to **recover on a blow-up**.

9.2 Funded cycle

Start: You now have a **\$200k funded account**.

You run a sequence of trades where **every trade on the prop is mirrored with an opposite hedge on your personal account**, sized so that:

- **Per 1% prop loss: hedge makes \$500**
- **Per 1% prop gain: hedge loses \$500**

Two main branches:

A. Funded fail before any payout

- Assume worst case: **10 SLs** at -1% each → -10% = -\$20,000 on prop
- Hedge:
 - $10 \times \$500 = \mathbf{+\$5,000}$

Compare with costToFunded = \$4,000:

- $5,000 - 4,000 = \mathbf{+\$1,000}$

So a funded fail **without any payout**:

- Recovers the **\$4,000 you paid to get funded**
- Leaves **\$1,000 net** profit

B. Funded payout hit

Target: **+5%** on prop = +\$10,000

- Your share at 90%: **\$9,000**
- Hedge:

- $5R \text{ loss} \times \$500 \text{ per } R = \textbf{-\$2,500}$

Net:

- $9,000 - 2,500 = \textbf{+\$6,500}$

You can potentially:

- **Hit several payouts** (each around **+\\$6,500 net**), and at some point,
- **Eventually blow the funded account**, which then triggers the **funded fail hedge logic**:
 - Recover costToFunded (4,000) + **extra 1,000 net**

So a **full super-cycle** (from first evaluation to eventual funded blow-up after multiple payouts) looks like:

1. Pay to get funded → structurally **recouped** via funded fail hedge.
2. Hit **one or more payouts** → each payout **~+\\$6,500 net**.
3. Eventually hit -10% DD on funded → **+ \$1,000 net** after recovering that 4k.

Over many repeats:

- **Eval cycles:**
 - Failures = +\$1,000 net each (in this model)
 - Passes = pay 4k (later recouped in funded fail)
- **Funded cycles:**
 - Payout cycles = +\$6,500 each
 - Blow-ups = +\$1,000 (and reset)

That's the **asymmetric engine** you've designed.

10 Final notes & risk warnings

Just to repeat clearly:

- This is a **structural, non-directional hedging framework**, not a prediction system.
- All numbers used are **exactly your numbers**:
 - 200k account
 - 10% / 5% phase targets
 - 5% daily DD, 10% max DD
 - \$1,000 evaluation fee
 - 1% risk per trade
 - Hedge yields of **\$200 per 1% eval loss** and **\$500 per 1% funded loss**
 - 90% profit split on funded payouts
- The entire logic is a **mathematical model** of how you want the money flows to work **if**:
 - Every hedge is perfectly mirrored
 - No slippage, no missed fills, no rule violations, no psychology issues
 - Prop firm rules and payouts are honoured exactly

 Not financial advice:

This does **not** guarantee profit, and it does **not** remove risk.

In reality, spreads, slippage, liquidity, execution errors, prop rules, and emotional decisions can all break this structure and cause losses on both accounts.